

HeartFortress – an Affective VR Game Adapting to Player Arousal

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ABSTRACT

This study explores affective computing in virtual reality (VR) gaming, focusing on adapting gameplay to players' arousal levels. By utilizing real-time physiological signals, such as heart rate and electrodermal activity, the VR game HeartFortress dynamically adjusts its mechanics. Observations suggest that the adaptive version of the game may enhance *flow* – a state of deep involvement in an activity – and increase physiological responsiveness, indicating heightened arousal. While preliminary, these findings highlight the potential of affective-driven adaptations in interactive systems, offering insights for gaming, training, and therapeutic applications.

Index Terms: Affective gaming, virtual reality, arousal.

1 INTRODUCTION

Video games have become a versatile medium capable of evoking a wide range of emotional responses. These emotions, whether satisfaction from strategic victories, excitement or fear from dynamic action, or empathy through storytelling, play a pivotal role in player engagement. However, the balance between challenge and engagement is critical. Overly simple games might lead to boredom, while excessively difficult ones can induce frustration, hindering the player's experience [9].

Affective computing offers the potential to design systems that recognize and respond to users' internal states, thereby enhancing human-computer interaction [7]. In gaming, this approach has birthed affective video games, which adjust gameplay in real time based on the player's physiological and behavioral responses allowing interpreting internal states [4]. By leveraging this adaptability, games can maintain an optimal state of *flow*, wherein a player is fully immersed, balancing challenge with skill [1].

Technological advancements now enable the collection and interpretation of biomedical data such as heart rate (HR) or electrodermal activity (EDA). Smartbands, smartwatches and similar devices allow developers to integrate these physiological signals into interactive applications, providing a foundation for adaptive systems. It has profound implications for personalizing the gaming experience, ensuring engagement by dynamically adjusting difficulty levels and other game parameters [10].

Virtual Reality (VR) has further transformed gaming by immersing players in fully interactive three-dimensional environments. By combining head-mounted displays and motion controllers, VR allows players to experience digital worlds in a manner that feels tangible and responsive, even if not including realistic graphics [5]. The immersive nature of VR enhances presence [5], making it a key platform for affective gaming. While VR can amplify emotional responses, such as arousal, this may indicate either engagement or stress depending on context and individual differences [8]. Recent developments in VR technology, including popular devices like the Meta Quest, have made these experiences accessible, enabling researchers to investigate how adaptive systems responding to real-time internal states can enhance user satisfaction [6].

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This study integrates principles from emotion theories, including work on universal emotions [2] and circumplex model of affect [8] (with valence denoting pleasantness and arousal for activation), to create an affective game environment. We examine how physiological feedback can inform dynamic difficulty adjustment, contributing to a more immersive player experience. By combining affective computing with gaming, the work aims to deepen understanding of how affective states influence user experience.

2 HEARTFORTRESS

We designed a first-person VR game, HeartFortress, where players defend a castle against waves of approaching enemies. Set within a 3D environment, players are stationed atop a tower equipped with a cannon. The objective is to prevent enemies from breaching the castle gates by aiming and firing at them. The gameplay spans 16 waves of increasing difficulty, with variations in enemy speed, spawn frequency, and pathways. Developed using the Unity engine (2021.3.5f1 LTS), the game incorporates assets such as the RPG Poly Pack Lite and 3D Cartoon Village to create a visually appealing world. Audio elements from Freesound, processed in Audacity, enhance gameplay with sound cues signaling events like wave starts and enemy actions.

Designed for the Meta Quest platform, HeartFortress incorporates VR-specific features such as intuitive motion controller aiming, real-time head and hand tracking, and a graphical interface that displays critical gameplay information, including *castle health*, wave progress, and weapon readiness. Unity's XR Interaction Toolkit and Oculus XR Plugin were used to support these features, ensuring smooth integration with the VR platform.

HeartFortress incorporates a real-time affective loop mechanism, driven by data acquired through the BITalino (r)evolution kit, which measures players' EDA and HR. This data dynamically adjusts gameplay mechanics to match the player's physiological state. For example, during periods of heightened HR, enemy movement speeds are gradually reduced using a logistic mapping function to help the player regain control. Conversely, if the EDA signal indicates reduced arousal, the system increases the frequency of enemy spawns and expands their spawn points to sustain player attention. Audio levels are adjusted, with lower values of EDA triggering louder sounds to encourage a physiological response.

The physiological feedback system begins with a calibration phase, during which baseline values for HR and EDA are established. Subsequent adjustments are computed by comparing real-time data against these baselines to alter game parameters. The calibration ensures that these adjustments are tailored to each individual player, accounting for natural physiological variability.

Two versions of the game were developed: one featuring this affective loop and the other with fixed, pre-defined mechanics. These versions allow for preliminary observations on how physiological responsiveness may influence player experience.

3 PROCEDURE

The testing of the game involved 10 participants, including six women and four men, aged between 24 and 56, with a median age of 25 years. Sixty percent of participants had little to no gaming experience, while one played once a month, two played twice a week, and one played daily. All participants were fully informed about the study's purpose and procedures and provided written informed consent before participating.



Figure 1: HeartFortress environment.

The experimental protocol included playing two versions of the game: one utilizing the affective feedback loop and another with static, predefined mechanics. To minimize order effects, half the participants started with the affective version, while the other half began with the non-affective version. Physiological data, specifically heart rate and electrodermal activity, were collected using the BITalino (r)evolution kit during gameplay, with sensors placed on the chest (HR) and fingers (EDA). Post-gameplay, participants completed questionnaires assessing their experience, including challenge, *flow*, competence, difficulty level and emotional engagement, based on the Game Experience Questionnaire (GEQ) framework [3] and answered questions regarding mechanics. The entire procedure, including setup, gameplay, and survey, was designed to last around 40 minutes.

4 RESULTS

The GEQ scores showed that the affective version of HeartFortress led to higher levels of *flow* ($M = 4.4$, $SD = 0.52$) compared to the static version ($M = 4.25$, $SD = 0.46$). Irritation was slightly lower in the affective version ($M = 1.4$, $SD = 0.22$) than in the static version ($M = 1.55$, $SD = 0.28$), while self-competence scores were somewhat lower in the affective version ($M = 3.8$, $SD = 0.49$) compared to the static version ($M = 4.3$, $SD = 0.48$). Challenge and difficulty levels were reported as higher in affective version ($M = 3.25$, $SD = 0.16$ vs. $M = 3.15$, $SD = 0.21$ and $M = 2.9$, $SD = 0.4$ vs. $M = 2.8$, $SD = 0.34$, respectively). Positive ($M = 4.35$, $SD = 0.48$) and negative affect ($M = 4.45$, $SD = 0.53$) remained the same. All results are presented as means (M) and standard deviations (SD).

Survey responses revealed that 9 out of 10 participants noticed differences in the number of enemies between the affective and static versions, and 7 out of 10 noticed variations in the speed of enemies. Changes in sound dynamics were observed by 7 participants. Most participants found the adaptive mechanics of the affective version more engaging, and only one participant reported that the physiological monitoring equipment interfered with gameplay.

Physiological data collected via the BITalino (r)evolution kit included HR and EDA. EDA data was mapped to a range of 0–25 μ S, and fluctuations in the signal were analyzed to identify arousal levels. The affective version showed variation in both HR and EDA, with noticeable spikes during gameplay for the latter signal. In the static version, the signals remained more stable.

5 DISCUSSION

These preliminary results suggest a potential for affective gaming systems like HeartFortress to enhance player engagement and immersion. The higher mean scores in *flow* for the affective version suggest that its adaptive mechanics might help balance challenge and engagement, contributing to a potentially more dynamic gaming experience. However, the slightly lower self-competence scores

could mean that while players found the adaptive mechanics engaging, they might have felt somewhat less confident in their abilities due to the dynamic adjustments in difficulty. The standard deviations across GEQ scores were relatively small, which might indicate that these effects were consistent and not driven by outliers.

Survey responses appear to support these findings, as most participants seemed to notice changes in enemy behavior and sound dynamics in the affective version. Encouragingly, the physiological monitoring equipment was reported to have minimal impact on gameplay, indicating its feasibility for research purposes and potential adaptability for consumer gaming. Physiological data from BITalino seemed to align with these subjective findings, showing increased HR variability and EDA spikes during the affective gameplay, which might reflect heightened arousal, while the static version showed more stable responses, possibly indicating less dynamic stimulation. However, specific in-game event data were not collected, making it impossible to directly analyze whether designed mechanics influenced arousal. Future studies should address these relationships to better evaluate their impact.

Beyond gaming, the methodology can extend to virtual training, mental health interventions, and education. Dynamic difficulty adjustments based on user stress or focus can enhance VR therapy and immersive learning. This work highlights affective computing's potential to create personalized, engaging experiences, driving innovation in interactive technology.

The small sample size of 10 participants limited the study's generalizability and precluded statistical analysis of differences between gameplay versions. Future research should include larger and more diverse participant pools to enable robust statistical comparisons. Additionally, while external physiological monitoring equipment was effective for this study, wearable technologies like smartwatches or VR-integrated sensors (e.g. enabling pupillometry) may provide more practical solutions for mainstream use. Expanding adaptive mechanics beyond enemy behavior and sound dynamics to include narrative changes or cooperative gameplay could further enhance arousal. Future studies could also incorporate self-reported arousal questionnaires to investigate the relationship between subjective arousal and physiological data. Lastly, the focus on short-term experiences leaves questions about long-term effects, warranting broader and iterative testing to refine affective gaming systems.

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